

Artificial Intelligence & Machine Learning

Task 3: Model Validation, Overfitting Control & Hyperparameter Tuning

California Housing Dataset | Intern Report

1. Overfitting Analysis

Overfitting occurs when a model learns the training data too well, including its noise, and fails to generalize to new unseen data. The key signal is a large gap between training error and test error.

Untuned Decision Tree Results:

The Decision Tree was first trained with no constraints (no max_depth limit):

- Train RMSE: ~0.0000 (model memorized training data perfectly)
- Test RMSE: ~0.7300 (model failed badly on new data)
- Gap: ~0.73 — this is a clear sign of severe overfitting

This happens because an unconstrained Decision Tree keeps splitting until every training sample is perfectly classified. It memorizes instead of learning patterns.

2. Cross-Validation Results

Instead of relying on a single train-test split (which can be lucky or unlucky), 5-Fold Cross-Validation was used. This splits the data into 5 parts, trains on 4, tests on 1, and repeats 5 times. The average gives a reliable performance estimate.

5-Fold Cross-Validation RMSE Scores:

Model	CV RMSE (5-Fold)	Interpretation
Linear Regression	0.7459	Stable, consistent performance
Ridge Regression	0.7459	Same as Linear — regularization had little effect
Decision Tree (Untuned)	0.9029	Worst — clear overfitting confirmed

Why cross-validation results are trusted: Each fold uses a different portion of data for testing. This removes dependence on any single split and gives a realistic estimate of how the model will perform on completely new data.

3. Hyperparameter Tuning (GridSearchCV)

GridSearchCV was used to systematically find the best hyperparameters for the Decision Tree. It tried every combination in the grid below, using 5-fold CV to evaluate each one (12 combinations x 5 folds = 60 total fits).

Parameter Grid Searched:

- max_depth: [3, 5, 7, 10] — controls how deep the tree can grow
- min_samples_split: [2, 5, 10] — minimum samples needed to split a node

Best Parameters Found:

- max_depth = 10
- min_samples_split = 10
- Best CV RMSE = 0.6366

Setting max_depth=10 prevents the tree from growing infinitely deep. Setting min_samples_split=10 ensures a node only splits if it has at least 10 samples, preventing the model from creating splits based on just 1-2 noisy data points. Together, these constraints reduced overfitting significantly.

4. Final Model Comparison

Model	Test RMSE	R2 Score	Rank
Tuned Decision Tree	0.6454	0.6821	1st (Best)
Ridge Regression	0.7456	0.5758	2nd
Linear Regression	0.7456	0.5758	3rd

5. Final Model Selection Justification

Selected Model: Tuned Decision Tree

Why this model was selected:

- Lowest Test RMSE of 0.6454 vs Linear Regression (0.7456) and Ridge Regression (0.7456)
- R2 score of 0.682 means the model explains 68.2% of the variance in house prices
- It outperforms linear models because house prices have non-linear relationships with features like location and income — which Decision Trees can capture

How overfitting was reduced:

- Before tuning: Train RMSE ~0.00, Test RMSE ~0.73 — gap of 0.73 (severe overfitting)
- After tuning: Train RMSE and Test RMSE gap reduced significantly
- max_depth=10 and min_samples_split=10 prevented the tree from memorizing training data

Why cross-validation results are trusted:

- 5-fold CV was used instead of a single split
- The model was tested on 5 different held-out portions of data
- CV RMSE of 0.6366 closely matches test RMSE of 0.6454 — confirming reliable generalization

Trade-offs considered:

- Linear Regression: simpler, faster, more interpretable — but cannot capture non-linear patterns
- Ridge Regression: adds regularization over Linear — but still constrained to linear relationships
- Tuned Decision Tree: more complex but achieves better accuracy with controlled overfitting — best choice for a production house price predictor where accuracy matters most